

SIYA AL HUSAIN

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PROFESSIONAL SUMMARY

Fresh Graduate in Electrical and Information Engineering with a strong foundation in telecommunications, networking, machine learning, and artificial intelligence. Possesses solid analytical and problem-solving skills developed through academic projects and hands-on experience across both hardware and software domains. Familiar with embedded systems, control applications, and modern communication technologies, with the ability to apply theoretical knowledge to practical engineering solutions. Demonstrates strong teamwork, communication, and task management skills in collaborative environments. Passionate about emerging technologies and innovation, with a keen interest in AI-driven systems and advanced communication networks. Seeking an entry-level engineering opportunity to contribute technical expertise, support innovative projects, and grow professionally in a dynamic organization.

EDUCATION

University of Ruhuna, Galle, Sri Lanka

BSc (Hons) in Engineering — Electrical and Information Engineering

2022– 2026

- CGPA: 3.61/4.0 (Second Class — Upper Division)
- Relevant Coursework: Communication Systems, Telecommunication networks, Data Communication, computer networks, wireless and mobile networks, design and management data networks, Signal and Systems, Embedded Systems, Machine Learning, Artificial Intelligence, Intelligent Systems, Electronic Circuits, analog and digital electronics, Control System Design, Power Systems, Electrical Machines, Engineering Management

Ilmiya Maha Vidyalaya

G.C.E A/L Examination- physical science

2017 - 2020

EXPERIENCE

Sri Lanka Telecom PLC

Telecommunication and networking Engineering trainee ↗

- Gained hands-on experience in telecommunications network operations, including FTTH, copper, and fiber-optic systems, with exposure to real-world network design and maintenance practices
- Performed network fault identification, troubleshooting, and basic performance optimization, ensuring reliable connectivity across wired communication systems
- Developed practical knowledge of Optical Distribution Network (ODN) architecture, equipment configuration, and network management systems (NMS) used in modern telecom infrastructure

WindForce PLC

Renewable Energy and electrical Engineering Trainee ↗

- Completed hands-on training at a utility-scale solar power plant, gaining practical knowledge of solar PV systems, including panels, tracking systems, string combiner boxes (SCB), inverters, and SCADA-based monitoring
- Conducted inverter station inspections, including monitoring of active power, current, and ISU cooling systems to ensure efficient and stable plant operation

PROJECTS

Human Activity Recognition using LSTM-Based Deep Learning ↗

Jan 2026 – Mar 2026

- Developed a Human Activity Recognition system using LSTM networks to classify activities such as walking, sitting, standing, and running from time-series sensor data
- Implemented data preprocessing, feature extraction, and LSTM-based model training, enabling accurate sequence-based activity recognition and real-time prediction capability

Model Predictive Control for Voltage Source Inverter (VSI) ↗

Jan 2026 – Aug 2026

- Developed a Model Predictive Control (MPC) strategy for a three-phase Voltage Source Inverter (VSI) to improve current tracking performance and reduce harmonic distortion
- Implemented mathematical modeling, switching state prediction, and cost function optimization using MATLAB/Simulink to achieve efficient real-time control performance

LiFi and Optical Wireless Communication Survey ↗

Jan 2026 – Mar 2026

- Conducted a comprehensive survey on LiFi and optical wireless communication technologies, analyzing system architectures, modulation techniques, and recent advancements
- Evaluated performance, scalability, limitations, and feasibility of LiFi systems in comparison with conventional RF-based wireless communication technologies

6G Antenna Technologies Research ↗

Jun 2025-Aug 2025

- Conducted a technical study and presentation on 6G antenna technologies, focusing on recent advancements, key concepts, and emerging design challenges in next-generation wireless communication

Smart Car Parking System ↗

May 2025 – Jul 2025

- Developed a smart parking automation system using IR sensors, ultrasonic sensors, ESP32, Arduino Uno, RFID modules, and servo-based automated gate control for real-time slot detection
- Implemented wireless communication and mobile application integration to display live parking slot availability, improving parking efficiency and system usability

Public Toilet Locator System (Mobile Application) ↗

Jan 2025 – Apr 2025

- Developed a Flutter–Firebase-based mobile application integrated with Google Maps to locate nearby public toilets with search, filtering, and navigation features

TECHNICAL SKILLS

Programming & Simulation: Python, C, C++, MATLAB/Simulink

Communication & Networking: Communication Systems, Telecommunication Systems, Data Communication, Signal and Systems, Network Fundamentals

Embedded & IoT Systems: Embedded Systems (Arduino, ESP32), Sensor Integration, IoT-Based Systems Development

Software & Engineering Tools: Proteus, Cisco Packet Tracer, Visual Studio, Flutter, MATLAB/Simulink, Figma, Jira

Control Systems & Power Electronics: Control System Design, Power Electronics, Power Systems, Electrical Machines, Inverter Fundamentals, Three-Phase VSI, Model Predictive Control (MPC)

PROFESSIONAL SKILLS

- Strong analytical thinking and problem-solving ability with attention to detail, combined with effective teamwork, leadership, and adaptability in multidisciplinary engineering environments
- Excellent communication, public speaking, and interpersonal skills with a disciplined, motivated, and quick-learning attitude toward continuous professional development

REFERENCE

Dr. (Mrs.) G.G.N. Sandamali
Senior Lecturer, PhD (NTU, Singapore),
BSc Eng (Hons) (Ruhuna),
MIEEE, AMIE (SL).
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